

Lecture 5

Uncertainty and Time Dependence

Lenny: *Good evening, General. Nice to see you again*.

The General: *Lenny? Is that you? It's been forever. Well, a long time anyway. Who's your friend?*

Lenny: *His name is Art. Art, shake hands with General Uncertainty*.

5.1 MATHEMATICAL INTERLUDE: COMPLETE SETS OF COMMUTING VARIABLES

5.1.1 States That Depend On More Than One Measurable

The physics of a single spin is extremely simple, and that's what makes it so attractive as an illustrative example. But that also means there's a lot it can't illustrate. One property of a single spin is that its state can be fully specified by the eigenvalue of a single operator, say σ_z. If the value of σ_z is known, then no other observable—such as σ_x—can also be specified. As we have seen, measuring either of these quantities destroys any information we may have had about the other one.

But in more complicated systems, we may have multiple observables that are compatible; that is, their values can be known simultaneously. Here are two examples: